

Qi-Hang Sun | Curriculum Vitae

🌐 <https://github.com/XZhoou> 🌐 <https://xzhoou.top>

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RESEARCH INTERESTS

Transportation network analysis (TNA), Travel demand management (TDM), Pedestrian dynamics

EDUCATION

University of Passau <i>Visiting Ph.D. Student</i>	Passau, DEU 2026.07 - Present
Beihang University <i>Ph.D. Candidate in Management Science and Engineering; GPA: 3.9/4.0; Rank: 2/51</i> Doctoral Thesis Topic: <i>Design and operational optimization of reservation schemes for congestion management</i>	Beijing, CHN 2023.09 - Present
Beihang University <i>B.S. in Biomedical Engineering (Minor)</i>	Beijing, CHN 2020.09 - 2023.06
Beihang University <i>B.S. in Information Management and Information System</i>	Beijing, CHN 2019.09 - 2023.06

AWARDS & HONORS

- Joint-training Ph.D. Program Scholarship, China Scholarship Council, 2025
- Excellent Student Cadre (Graduate), Beihang University, 2025
- The First Prize Academic Scholarship (Graduate), Beihang University, 2024, 2025
- Merit Student (Graduate), Beihang University, 2024
- Excellent League Member (Graduate), Beihang University, 2024, 2025
- Best Paper Award, The 5th Frontier Symposium on Traffic Behavior and Transportation Science, 2024
- Freshmen Scholarship, Beihang University, 2023
- Outstanding Undergraduate Graduates, Beihang University, 2023
- Merit Student (Undergraduate), Beihang University, 2022
- Model Student of Academic Records, Beihang University, 2022
- The Grand Prize Academic Scholarship (Undergraduate), Beihang University, 2022
- The First Prize Discipline Competition Scholarship, Beihang University, 2022
- The Second Prize Social Work Scholarship, Beihang University, 2022
- Honorable Mention Prize in Mathematical Contest In Modeling (MCM), COMAP, 2022
- The Second Prize in China Undergraduate Mathematical Contest in Modeling, CUMCM, 2021
- The Second Prize Academic Scholarship (Undergraduate), Beihang University, 2021
- Outstanding Volunteer, Beihang University, 2021

INTERNSHIP EXPERIENCE

Ministry of Industry and Information Technology Undergraduate Research Assistant	2022.03-2022.09
International Relations Department Student Assistant	2025.09-2026.01

SKILLS

Programming	Python, R, Matlab, C, SQL, $\text{\LaTeX}$
Languages	Chinese (<i>Native</i>), English (<i>Conversational</i>)
Software	MS Office, IBM SPSS
Sports	Football, Badminton, Table tennis

PUBLICATIONS

- **Reservation scheme for congestion management during morning commute: Considering noncompliance**
Submitted to Transportation Research Part C: Emerging Technologies (2nd Round Revision)
Qi-Hang Sun, Ren-Yong Guo
- **Reservation-based congestion management under noncompliance: Auction, reallocation, and penalty**
Working paper
Qi-Hang Sun, Ren-Yong Guo
- **Noncompliance as a Double-Edged Sword in Reservation Systems: Equilibrium Analysis and Adaptive Design**
Working paper
Qi-Hang Sun, Ren-Yong Guo
- **Quantifying reservation-scheme effectiveness and noncompliance-induced efficiency loss: Evidence from a large-scale morning-commute simulation**
Working paper
Qi-Hang Sun, Ren-Yong Guo
- **When drones meet elevators: Managing vertical congestion through drone-elevator coordination in high-rise buildings**
SSRN Preprint: 7119618
Qi-Hang Sun, Shixin Zhao, Ren-Yong Guo

CONFERENCE TALKS

- **The 14th Workshop on Computational Transportation Science**, Shanghai, 2023.08
Oral Presentation
- **The 5th Frontier Symposium on Traffic Behavior and Transportation Science**, Yichang, 2024.08
Oral Presentation
- **The 16th Workshop on Computational Transportation Science**, Wuhan, 2025.07
Oral Presentation
- **The International Symposium on Multimodal Transportation**, Guangzhou, 2025.11
Oral Presentation
- **Seminar Series 2025/2026, Future Cities Lab**, Online, 2026.04
Oral Presentation